

# Adder circuit

Condensed Study Notes

Generated: 2026-07-17 19:43 UTC

## Half Adder Details

The half adder is a simple electronic integrated circuit used for addition in two binary values.

- It performs binary addition with the addition of a single bit. It's fundamental in building higher-level circuits.
- Inside, it has three inputs - A and B, and two outputs –  $\Sigma$  (the sum) and C (carried-out). The carried out bit is used to indicate whether carry-overs were involved during addition.

Think of it like this: adding two numbers by hand; each digit in the result corresponds to one of the inputs. It's crucial for building the foundation of binary operations, because its output defines how a larger operation will proceed.

## COMPUTER ARCHITECTURE ADDER CIRCUIT

[Content could not be generated for this section.]

## Full Adders - The Core Logic

<<When an adder needs to add two bits together, it uses a full adder. It combines the two single bits with a carry-in from the previous stage into one overall output. This allows you to make calculations on multiple bit values using a single circuit. It's like adding apples – each one is slightly separate but when combined all at once results in a significant piece.

## Understanding Adders – The Basic Core

### Adders Build Numbers from Bits

The fundamental function of an adder is to combine multiple bits into a single, more complex number.

A digital circuit adds corresponding bits together using a control that provides feedback to guide the summing process.

### The Basics

- An Adder consists of two inputs, usually labeled as A and B, and a carry-in stage. The carry-in acts like an additional input - it provides info about intermediate results when adding more bits from the previous stage.